

Sizing a Rooftop Solar Array

Finding the area of an oblique triangle from survey bearings

Group Monday Righteous

Trigonometry Group Activity

August 16, 2026

Outline

- 1 The problem
- 2 From bearings to an angle
- 3 Finding the area
- 4 Discussion

The brief

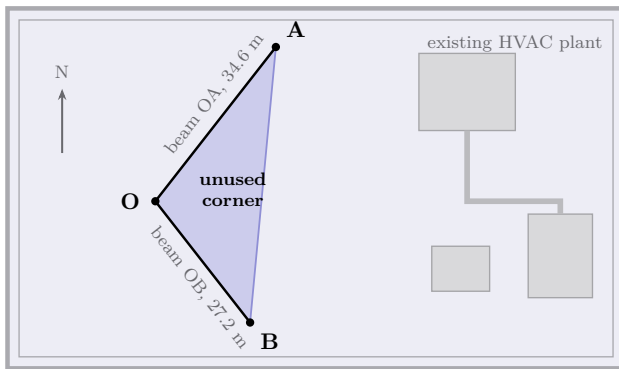
An engineering firm is scoping a small solar array in an unused triangular corner of a warehouse rooftop, boxed in by two existing structural beams. They need **the enclosed area**, which is what sets how many panels the site can hold.

The catch

The surveyor never recorded an angle. The field notes contain **two bearings**, which is how rooftop and land surveys are actually reported.

So the job splits in two: recover the angle, then find the area.

The site



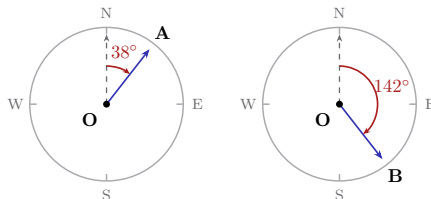
Plan view of the warehouse rooftop, bounded by two structural beams

What the surveyor actually recorded

Field notes at station O

Beam OA	34.6 m, bearing 38°
Beam OB	27.2 m, bearing 142°
Angle at O	<i>not recorded</i>

A total station measures **distance** and **direction**, and direction is logged as a bearing. Nobody stands in the corner with a protractor, so **the included angle has to be derived**.



A bearing is always measured *clockwise from north*, from the same origin each time.

From two bearings to one angle

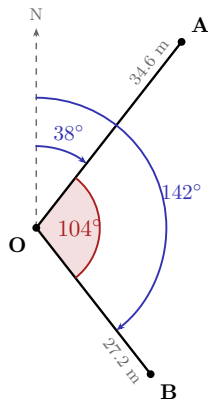
Two conditions make this a plain subtraction:

- both bearings share the **same reference point**, O;
- both are measured **clockwise from north**.

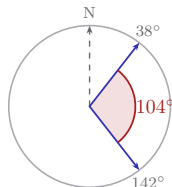
Included angle

$$\angle AOB = 142^\circ - 38^\circ = 104^\circ$$

No trigonometry yet. This step is bookkeeping about how the directions were recorded.

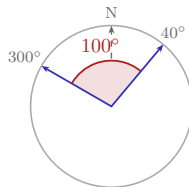


The rule in general



$$142 - 38 = 104 < 180$$

take it directly



$$300 - 40 = 260 > 180$$

use $360 - 260 = 100^\circ$

Subtract the bearings; if the result exceeds 180° you have measured the *reflex* side, so take 360° minus it instead. Here $104^\circ < 180^\circ$, so no adjustment is needed.

Now it is a standard SAS triangle

Known

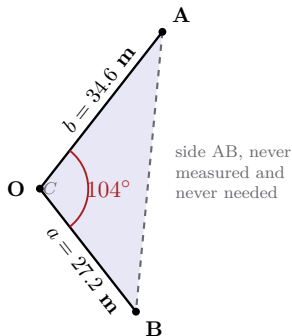
$$a = OB = 27.2 \text{ m}$$

$$b = OA = 34.6 \text{ m}$$

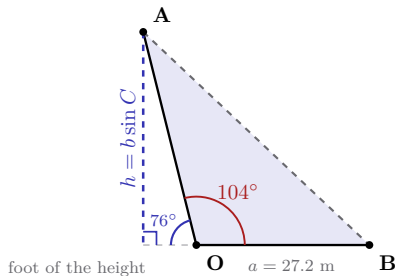
$$C = 104^\circ \text{ (derived)}$$

The angle sits **between** the two known sides, which is exactly the SAS pattern.

C is obtuse, so the triangle is genuinely oblique: no right angle to fall back on, and no perpendicular height in the field notes.



Why $\frac{1}{2}ab \sin C$ is the right tool



$$\begin{aligned}\text{Area} &= \frac{1}{2} \times \text{base} \times \text{height} \\ &= \frac{1}{2} a (b \sin C)\end{aligned}$$

C is obtuse, so the foot lands outside the triangle, and $\sin 104^\circ = \sin 76^\circ$ still holds.

The familiar $\frac{1}{2} \times \text{base} \times \text{height}$ never went away. $\sin C$ simply supplies the height from the side and angle we do have.

The calculation

$$\begin{aligned}\text{Area} &= \frac{1}{2} ab \sin C \\ &= \frac{1}{2} \times 34.6 \times 27.2 \times \sin 104^\circ \\ &= 470.56 \times \sin 104^\circ \\ &= 470.56 \times 0.9703 \\ &\approx 456.58 \text{ m}^2\end{aligned}$$

Watch for

Calculator in **degrees**, not radians.

$\sin 104^\circ = \sin 76^\circ = 0.9703$. An obtuse angle gives a perfectly ordinary positive sine.

Round only at the **final step**.

Complete solution

Quantity	Value
Side OA (b)	34.6 m
Side OB (a)	27.2 m
Bearing to A	38°
Bearing to B	142°
Included angle C	$142^\circ - 38^\circ = 104^\circ$
$\sin 104^\circ$	≈ 0.9703
$\frac{1}{2} \times \text{OA} \times \text{OB}$	470.56
Area	$\approx 456.58 \text{ m}^2$

The answer

Area of the triangular rooftop section

$$\text{Area} = \frac{1}{2} \times 34.6 \times 27.2 \times \sin 104^\circ \approx 456.58 \text{ m}^2$$

From two beam lengths and two bearings, with no angle ever measured.

A note on rounding

$\sin 104^\circ = 0.970296\dots$, so

$$470.56 \times 0.970296 = 456.582\dots$$

A rounded 0.9709 would give
 456.87 m^2 , about 0.3 m^2 adrift,
roughly **one panel's worth** of area.

Small, but the kind of error that
compounds: carry the full value
through and round once, at the end.

Sensitivity of the answer

$$\sin C = 0.9703 \quad 456.58 \text{ m}^2$$

$$\sin C = 0.9709 \quad 456.87 \text{ m}^2$$

$$\text{difference: } 0.29 \text{ m}^2$$

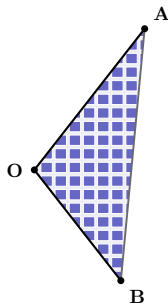
Optional: what could that area produce?

Utility-scale arrays are often quoted at roughly 170 W of installed capacity per square metre:

$$456.58 \times 170 \approx 77\,619 \text{ W} \approx 77.6 \text{ kW}$$

Say this out loud

A rough, order-of-magnitude estimate, not an engineering calculation. It ignores tilt, row spacing, shading and inverter losses, all of which reduce the real figure.



schematic only, ignoring tilt, row spacing and shading

What to take away

1. The angle was earned, not given

Bearings share a reference direction and a rotational sense, so their difference *is* the angle between the lines. Real survey data arrives this way; converting it is part of the problem.

2. It is still just base $\times$ height

$\frac{1}{2}ab \sin C$ is not a new formula to memorise. $\sin C$ stands in for a perpendicular height that was never measured, which is why SAS is enough.

From two distances and two directions to 456.58 m^2 , and a panel count.

That's all!